

Term Test II

Subject: Data Science and Analytics

Date	08/11/21	Full Marks	50
Level	III	Time	1.5 hrs
Programme	BCI		
Semester	VII		

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

- What is multivariate linear regression? Describe the OLS estimation techniques for the multi-variate regression with numerical examples. [7]
 - Describe the Nadaraya-Watson kernel regression estimator for the conditional mean $E[Y|X=x]$. Clearly show how the formula is obtained using kernel weighting. Explain the role of the bandwidth in kernel regression and discuss strategies for selecting an optimal bandwidth. [8]
- How can we be sure that an event A causes an event B? Differentiate correlation and causation with an appropriate example. [7]
 - Explain logistic regression and its application in classification problems. [8]
- Suppose that you are estimating a linear model.

$$y = a_0 + a_1x_1 + a_2x_2 + a_3x_3$$
 Now, you decided to calculate the variance inflation factor for the variable x_3 . Following the usual procedure, you regressed x_3 on x_1 and x_2 with the results summarized in the following table.

x_1	x_2	x_3	Fitted x_3
1.0	2.0	6.2	6.5
2.0	1.5	6.4	6.75
1.5	2.5	8.9	8.75
2.5	2.0	8.6	8.75
3.0	1.0	7.0	7.0
1.2	1.8	6.0	6.3
2.2	2.2	9.5	9.35
1.8	2.0	8.0	8.1
2.8	1.5	8.2	8.25
2.0	2.5	9.9	9.75

- What is the variance inflation factor for x_3 ? [8]
- What is clustering? Describe k-means and distance metrics for the mixed data types with numerical examples. [7]
- Write short notes: (Any two) (2.5 * 2 = 5)
 - Decision Trees and the CART algorithm
 - ARIMA models
 - Granger causality